

Practical Membership Inference Attacks Against Large-Scale Multi-Modal Models: A Pilot Study

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Abstract

Membership inference attacks (MIAs) aim to infer whether a data point has been used to train a machine learning model. These attacks can be employed to identify potential privacy vulnerabilities and detect unauthorized use of personal data. While MIAs have been traditionally studied for simple classification models, recent advancements in multi-modal pre-training, such as CLIP, have demonstrated remarkable zero-shot performance across a range of computer vision tasks. However, the sheer scale of data and models presents significant computational challenges for performing the attacks.

This paper takes a first step towards developing practical MIAs against large-scale multi-modal models. We introduce a simple baseline strategy by thresholding the cosine similarity between text and image features of a target point and propose further enhancing the baseline by aggregating cosine similarity across transformations of the target. We also present a new weakly supervised attack method that leverages ground-truth non-members (e.g., obtained by using the publication date of a target model and the timestamps of the open data) to further enhance the attack. Our evaluation shows that CLIP models are susceptible to our attack strategies, with our simple baseline achieving over 75% membership identification accuracy. Furthermore, our enhanced attacks outperform the baseline across multiple models and datasets, with the weakly supervised attack demonstrating an average-case performance improvement of 17% and being at least 7X more effective at low false-positive rates. These findings highlight the importance of protecting the privacy of multi-modal foundational models, which were previously assumed to be less susceptible to MIAs due to less overfitting. Our code is available at <https://github.com/ruoxi-jia-group/CLIP-MIA>.

1. Introduction

Membership inference attack (MIA) is a type of privacy attack that attempts to determine if a specific data point was used to train a machine learning model [14]. This type of attack can compromise the privacy of individuals whose data was used to train the models [14], but can be also used to identify vulnerabilities, privacy leakage, and unauthorized use of personal data in machine learning models [21, 31].

Moreover, the rise of a foundational model trained on vast amounts of open data has highlighted the potential breach of contextual integrity, a fundamental principle in legal discussions of privacy [22]. MIA can, therefore, be used as an effective tool for individuals to check if companies store their personal information and request its deletion to comply with the European General Data Protection Regulation (GDPR).

Existing MIAs that achieve advanced performance rely on the idea of shadow training [4, 35]. The shadow models are usually at a scale of hundreds [4] and are with the same or similar architecture as the target model. The training algorithm needs to be the *same* as the one that trains the target model. The difference between shadow models that contain a certain sample and those that do not is then utilized to learn proper features to identify the membership of an individual sample. However, reliance on shadow training forecloses its application to large-scale models. As a concrete example, training Contrastive Language-Image Pre-Training (CLIP) [23], a cutting-edge multi-modal learning paradigm, takes 18 days even with hundreds of advanced GPUs [10]. This makes shadow training to attack CLIP prohibitively expensive. In addition, obtaining full details of the training algorithm performed by these state-of-the-art models is often difficult, as they are considered intellectual property and are not published.

This paper is a **pilot study of practical MIAs against large-scale multi-modal models**. The proposed techniques bypass the shadow training and work with only black-box access. We use CLIP as an example for the extensive evaluation, due to the following reasons: First, CLIP is

widely used for zero-shot learning in various computer vision tasks [25, 24, 8], which entails the understanding of its privacy risk. Also, CLIP has been trained on massive data scraped from the Internet with undisclosed algorithms, thus exemplifying the scale and the threat model for other emergent foundation models [3]. Our contributions are summarized as follows.

(1) Benchmarking the susceptibility of CLIP to membership inference. We introduce a simple baseline attack that identifies membership based on cosine similarity (CS) between image and text features, akin to those MIAs designed for single-modal models based on loss or confidence scores in prior literature [37, 19, 32]. The rationale for using CS as a signal for membership inference is that CLIP is trained to maximize CS on training samples, which can result in members having higher CS than non-members. This simple method achieves a reasonable membership identification accuracy ranging from 66.5% to 78.8% but has limited performance in the low false-positive regime.

(2) Improving the CS-based attack via target data augmentations. We develop an enhanced MIA technique that involves applying transformations to a specific point and aggregating the changes in CS across various common transformations (e.g., resizing, cropping, rotation, and translation). Our evaluation indicates that this method consistently improves attack performance across multiple datasets and CLIP model architectures, albeit to a small extent. The inspiration for this technique stems from our empirical observations that training points experience a larger CS drop than non-training ones when subject to transformations.

(3) Developing a new weakly supervised MIA framework given one-sided non-member information. Both the baseline attack and the augmentation-enhanced attack demonstrate high performance without requiring ancillary information, but their accuracy is limited at false positive rates (e.g., less than 10% true positive rate at a false positive rate of 1%). In this paper, we identify a new threat model that is plausible for models trained on Internet data. Specifically, we consider a scenario where the attacker has *one-sided* knowledge about non-members. By scraping Internet data posted after the target model’s publication date, the attacker can acquire a set of data guaranteed not to have participated in the training. We propose a weakly supervised attack that utilizes this non-member set to construct a model that predicts membership. This approach shows remarkable attack performance, improving the baseline accuracy by 17% and being around 7X more effective at low false-positive rates, despite the absence of information about members.

(4) Exploring potential defenses. Our findings demonstrate the vulnerability of large-scale multi-modal models to membership inference risks. To address this issue, we

investigate potential defenses and their associated privacy-utility tradeoff. This pilot study serves as a starting point for assessing the privacy risks associated with emerging large-scale multi-modal models.

2. Related Work

Membership inference attacks (MIAs) are designed to determine whether a given data sample has been used to train a particular model. These attacks typically leverage the overfitting tendency of machine learning models, which often show varying predictive behaviors on training data versus unseen data. While most existing MIAs have been developed for small-scale classification models [14], where overfitting is more prevalent, this paper focuses on investigating MIAs against large-scale multi-modal models.

One approach taken by MIAs involves identifying a metric that can differentiate between the behaviors of member and non-member samples. Example metrics include prediction loss [37], correctness [19, 32], and entropy [30, 32]. Our baseline attack strategy is similar to the metric-based attack, adapted to the multi-modal setting.

Another popular approach [30] is through building an attack model, which takes the predicted confidence vector of the target model on a target sample as input and infers the membership of the sample as output. To obtain the “training data” for the attack model, one needs to perform shadow training, which involves training numerous models on different subsets of data. A point’s membership within a shadow model and the corresponding features extracted from that shadow model can then form a training pair for the attack model. Recent studies by [4, 35] have explored likelihood-ratio-based MIAs, which estimate the probability of having a certain loss for both members and non-members and then identify the membership based on the likelihood ratio. These approaches also require shadow training. The losses of the shadow models trained with and without the target point are used to estimate the likelihood ratio. Although these techniques achieve state-of-the-art attack performance, especially in the low false-positive regime, their reliance on shadow training makes them computationally infeasible for large-scale models. Additionally, shadow training assumes knowledge of the training procedure and target model architecture, which becomes problematic as many large models are only available as black-box APIs.

MIAs against text-to-image generation models [36] and image captioning models [15] have recently been proposed to keep pace with the increasing use of multi-modal models. While sharing a similar scope to our work, their considered training size is relatively small, and thus shadow training is still applicable. By contrast, we propose MIAs for concurrent large-scale multi-modal models for which shadow training is intractable. A recent paper [13] also

studies privacy risks associated with CLIP; however, unlike our paper which provides a general, application-agnostic attack method, their method can only be applied to inferring whether an identity’s face images are in the training set.

3. Threat Model

This paper focuses on designing MIAs against large-scale vision-language models trained via CLIP [23]. CLIP consists of an image encoder f_{img} and a text encoder f_{txt} . Given a training set $\mathcal{D}_{\text{tm}} = \{(x_i, y_i)\}_{i=1, \dots, n}$, where x_i represents the image portion of the i -th training sample and y_i represents the corresponding text portion. CLIP jointly trains the image encoder and the text encoder via contrastive learning. The training objective aims to maximize the CS between correct pairs of image and text features while minimizing the similarity between incorrect pairs.

We summarize the attacker’s goal and the assumptions on the attacker’s knowledge as follows.

Goal. Consider the training phase of a target model, samples in the training set are called *members*, and those not used for training are called *non-members*. For a target model, the goal of a membership inference attack (MIA) is to infer whether a given sample is a member or not.

Model Knowledge. We consider *black-box* access to the target model, where the attacker can query the model with pairs of images and corresponding text captions, and in return receive image and text features. In particular, the attacker does not know the architectural design and parameters of the target model or the training algorithm.

Data Knowledge. We explore two different settings regarding the attacker’s knowledge of the training data used for the target model. In the first setting, called the *zero-knowledge setting*, we assume that the attacker has no knowledge of the data distribution used to train the target model. In the second setting, which is similar to past research [4, 35, 29], we assume that the attacker has access to a set of samples $\mathcal{D}_{\text{all}}$ that represents the underlying data distribution $\mathbb{D}$. This assumption is based on the fact that large-scale multi-modal models are often trained on data collected from the internet, allowing the attacker to scrape data from the internet as a proxy for the underlying distribution. In this second setting, we further assume that a set $\mathcal{D}_{\text{no}}$ is known to be non-members, while the membership of $\mathcal{D}_{\text{all}}$ is unknown. For example, one can easily collect non-members by scraping the data posted after the model’s publication, but membership of data posted before the publication date is not discernible. This indicates that ground-truth membership information is *one-sided*: we can only obtain information about non-members precisely. More specifically, consider the open-CLIP models, which are trained

on LAION 400M. The authors have released information regarding their data collection: ‘The dataset has been extracted from random web pages crawled between 2014 and 2021.’ Furthermore, the original CLIP model was published in January 2021, and GPT-4 was officially launched on March 2023, training on data collected before September 2021. Therefore, it is realistic for prospective attackers to leverage the published date of the target model and create a non-member dataset by scraping the Internet data posted after the publication date of the target model. Alternatively, one can obtain a ground-truth set of non-members by keeping some data internal and not contributing it to the training of the target model. We refer to this second setting as the *one-sided-knowledge setting*.

Computational Resources. Our focus is on a practical attack scenario, where the attacker has limited computing resources and must rely on efficient attack strategies. Specifically, our goal is to entirely circumvent shadow training computations, which involve repeatedly re-training the target model.

4. Methodology

In this section, we introduce three attack strategies for large-scale multi-modal models. The first two strategies are applicable to the zero-knowledge setting, while the third strategy can take advantage of one-sided non-member information.

4.1. A Baseline: Cosine Similarity Attack

The key idea is that the target model is trained to maximize cosine similarity between image and text features on members, so the attacker will receive higher cosine similarity scores from members than from non-members. Given the cosine similarity between the input image x and the text y , we predict (x, y) as a member if $\text{CS}(f_{\text{img}}(x), f_{\text{txt}}(y)) > \tau$ for some threshold $\tau > 0$. We refer to this attack as the *Cosine Similarity Attack (CSA)*. This attack has a similar flavor to those that threshold classification confidence score or loss to infer data membership for classification models [37].

4.2. Augmentation-Enhanced Attack

One idea for achieving stronger attack performance is to leverage the data augmentation technique, inspired by [18, 7]. In particular, for multi-modal models, we observe that after applying transformations to each target sample, the decrease in cosine similarity is more significant for member samples than for non-member samples. This observation enables us to leverage the CS gap induced by transformations to improve the attack performance.

Formally, we consider K transformations $\mathcal{T} = \{T_1(\cdot), \dots, T_K(\cdot)\}$. For each transformation, we compute the CS associated with the transformed target sample,

namely $CS(f_{\text{img}}(T_k(x)), f_{\text{txt}}(y))$, and then calculate the gap from the CS associated with the original sample:

$$\Delta CS_k(x, y) := CS(f_{\text{img}}(x), f_{\text{txt}}(y)) - CS(f_{\text{img}}(T_k(x)), f_{\text{txt}}(y)). \quad (1)$$

To improve the baseline attack, we aggregate CS gaps over all transformations considered in addition to the CS of the original image, i.e., $CS(x, y) + \sum_{k=1}^K \Delta CS_k(x, y)$, and predict (x, y) as a member when this aggregate quantity exceeds a certain threshold. We refer to this method as the *Augmentation-Enhanced Attack (AEA)*.

4.3. Weakly Supervised Attack

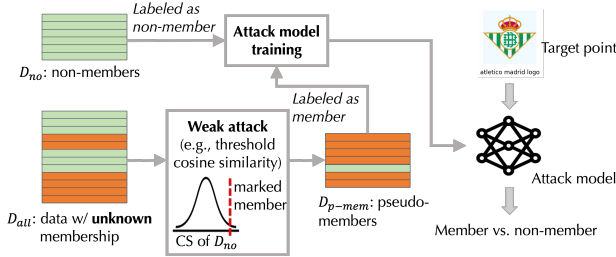


Figure 1. Overview of the WSA.

Different from the AEA, our third approach uses ground-truth information about non-members *in tandem with* noisy information about members provided by a *weak* attack method to learn an attack model. The attack pipeline is illustrated in Figure 1.

The attack starts by querying the target model for the image and text features associated with each non-member in $\mathcal{D}_{\text{no}}$. We denote the collection of non-member features as $F_{\text{no}} = \{(f_{\text{img}}(x), f_{\text{txt}}(y)) | (x, y) \in \mathcal{D}_{\text{no}}\}$. At the same time, we evaluate CS between the features: $CS_{\text{no}} = \{CS(f_{\text{img}}(x), f_{\text{txt}}(y)) | (x, y) \in \mathcal{D}_{\text{no}}\}$. Based on CS_{no} , we can estimate the distribution of CS scores for non-members. As we will show later, the distribution has a Gaussian shape. We can further approximate the mean and variance of the distribution through the sample mean and sample variance of CS_{no} , denoted by μ_{no} and σ_{no}^2 . Intuitively, a sample with CS significantly larger than μ_{no} (e.g., larger by $\lambda\sigma_{\text{no}}$ for some constant λ) is likely to be a member.

Now, our goal is to find a subset of $\mathcal{D}_{\text{all}}$ most likely to be members. To achieve this, we query the target model for image-text features associated with $\mathcal{D}_{\text{all}}$ and calculate the CS score for each sample. We compare the scores of each sample with the distribution of non-member scores characterized by μ_{no} and σ_{no} and further mark the samples with CS significantly larger than μ_{no} as members. We call the set of members marked based on CS the *pseudo-member set* because this set is likely to contain non-members as well. Formally, we represent the pseudo-member set as

$$\mathcal{D}_{\text{p-mem}} = \{(x, y) | CS(x, y) \geq \mu_{\text{no}} + \lambda\sigma_{\text{no}}, (x, y) \in \mathcal{D}_{\text{all}}\}. \quad (2)$$

Note that the pseudo-member set $\mathcal{D}_{\text{p-mem}}$ only provides noisy information about membership.

Given $\mathcal{D}_{\text{no}}$ and $\mathcal{D}_{\text{p-mem}}$, we can construct an *attack dataset* by associating each text-image feature derived from $\mathcal{D}_{\text{no}} \cup \mathcal{D}_{\text{p-mem}}$ with a label indicating membership. Specifically, the attack dataset is denoted by

$$\mathcal{D}_{\text{attack}} = \{(f_{\text{img}}(x), f_{\text{txt}}(y), b) | (x, y) \in \mathcal{D}_{\text{no}} \cup \mathcal{D}_{\text{p-mem}}, b = \mathbb{1}[(x, y) \in \mathcal{D}_{\text{p-mem}}]\}, \quad (3)$$

where b is binary with 1 representing a potential member and 0 representing a potential non-member.

Finally, using $\mathcal{D}_{\text{attack}}$, we can train an attack model f_{attack} that takes the image-text as input and returns its membership status. Once the attack model is trained, we can use it to attack any target sample (x, y) by first querying the target model for its features $(f_{\text{img}}(x), f_{\text{txt}}(y))$ and feeding the features at the input of f_{attack} to receive the membership inference result, namely, $f_{\text{attack}}(f_{\text{img}}(x), f_{\text{txt}}(y))$.

5. Evaluation

In this section, we thoroughly evaluate the performance of our attacks (i.e., CSA, AEA, and WSA) across various models and datasets and explore potential defenses.

5.1. Evaluation Setup

Datasets. To evaluate our attack, we use the mixture of LAION [26], Conceptual Captions 3M (CC3M) [27], Conceptual Captions 12M (CC12M) [5], and MSCOCO [20] datasets as $\mathcal{D}_{\text{all}}$. These datasets are all in web-data format (e.g., image URLs and corresponding text captions) and can be scraped using the technique in [2]. LAION, CC3M, CC12M, and MSCOCO contain approximately 400M, 3.3M, 12M, and 600K image-text pairs, respectively. In particular, the LAION dataset is used by the open-sourced CLIP (OpenCLIP) implementation that we consider in this paper. We use one of the datasets to train a target model and sample from the combination of the other three datasets to get the ground-truth non-members. We further split the non-members into two disjoint subsets: one provides the ground-truth non-member information for the WSA, while the other is used for evaluating the attack performance.

Note that our setup of using different open-world datasets as members and non-members differs slightly from the traditional MIA evaluation setting focused on small-scale classification models, where the same dataset is split into two subsets as members and non-members. In the main paper, we did not adopt the traditional setting for two reasons. Firstly, all the datasets considered are open-world data collected from the Internet; our setting, where the model owner uses one dataset to train the model and the attacker

Table 1. Attack performance evaluation across different target model architectures and datasets. Δ indicates the improvement from CSA.

LAION													
Dataset													
Model	ViT-B/32				ViT-B/16				ViT-L/14				Runtime
Metric	AUC	Δ	TPR@1%FPR	Δ	AUC	Δ	TPR@1%FPR	Δ	AUC	Δ	TPR@1%FPR	Δ	(Sec)
CSA	0.7460 $\pm$ 0.0012	-	0.0723 $\pm$ 0.0050	-	0.7593 $\pm$ 0.0117	-	0.0758 $\pm$ 0.0004	-	0.7876 $\pm$ 0.0031	-	0.0487 $\pm$ 0.0037	-	670.7s
AEA	0.7625 $\pm$ 0.0012	0.0165	0.0940 $\pm$ 0.0004	0.0217	0.7897 $\pm$ 0.0003	0.0304	0.0920 $\pm$ 0.0005	0.0162	0.7950 $\pm$ 0.0025	0.0074	0.0836 $\pm$ 0.0023	0.0349	830.1s
WSA	0.9199 $\pm$ 0.0009	0.1739	0.7212 $\pm$ 0.0069	0.6489	0.9349 $\pm$ 0.0044	0.1756	0.7381 $\pm$ 0.0061	0.6623	0.9413 $\pm$ 0.0026	0.1537	0.7611 $\pm$ 0.0104	0.7124	1539.2s
CC12M													
Dataset													
Model	RN50				RN101				ViT-B/32				Runtime
Metric	AUC	Δ	TPR@1%FPR	Δ	AUC	Δ	TPR@1%FPR	Δ	AUC	Δ	TPR@1%FPR	Δ	(Sec)
CSA	0.6650 $\pm$ 0.0026	-	0.0224 $\pm$ 0.0011	-	0.6854 $\pm$ 0.0040	-	0.0361 $\pm$ 0.0011	-	0.6957 $\pm$ 0.0051	-	0.0395 $\pm$ 0.0013	-	670.7s
AEA	0.7447 $\pm$ 0.0029	0.0797	0.0948 $\pm$ 0.0039	0.0724	0.7306 $\pm$ 0.0000	0.0453	0.0930 $\pm$ 0.0020	0.0569	0.705 $\pm$ 0.0016	0.0093	0.0775 $\pm$ 0.0013	0.0380	830.1s
WSA	0.7945 $\pm$ 0.0066	0.1295	0.3724 $\pm$ 0.0138	0.3500	0.8171 $\pm$ 0.0031	0.1318	0.4032 $\pm$ 0.0066	0.3671	0.7931 $\pm$ 0.0006	0.0974	0.3306 $\pm$ 0.0058	0.2911	1539.2s

scrapes another dataset as non-member, simulates the real-world attack case. Secondly, the original OpenCLIP is trained on the *full* LAION dataset and we would like to attack this model to demonstrate the real-world impact of our attacks. In fact, we found that training OpenCLIP on smaller subsets of LAION (e.g., 240M) leads to significant performance degradation, thereby yielding a less interesting attack case.

Data Processing. For large-scale datasets scraped from the internet at scale, it is natural to have some overlapping (e.g., LAION has overlapped samples with other datasets). Therefore, we include the text preprocessing step and the URL processing step to remove the intersection so that $\mathcal{D}_{\text{no}} \cup \mathcal{D}_{\text{trn}} = \emptyset$ and the image-text pairs used to build $\mathcal{D}_{\text{attack}}$ has no overlap with those used to evaluate the attack performance. The details of the preprocessing step are provided in Appendix E.

Target Models. Since the original CLIP paper [23] does not disclose their training algorithm details, we utilize the reproduced OpenCLIP [6] as a surrogate, and it achieves performance comparable to what was reported in the original paper. For the LAION dataset, we primarily focus on a vision transformer-based architecture (ViT) [9], given its superior performance. We utilize ViT-B/32, ViT-B/16, and ViT-L/14 as our target pre-trained models, following [6]. We use a pre-trained ResNet50 (RN50) trained on the CC12M dataset and additionally train ResNet101 (RN101) and ViT-B/32 on the same dataset to evaluate our attack performance on different datasets and architectures. We will refer to ViT-B/32 and RN101 on CC12M as self-trained models, while the others will be referred to as pre-trained models. It is worth noting that ResNet structures are recommended for CC12M since vision transformers tend to be more data-hungry than ResNet [12].

Evaluation Metrics. We evaluate our approach based on three metrics. The first is the area under the curve (AUC)

score of the receiver operating characteristic (ROC) curve. This is an average metric that takes an average over all false-positive rates. The second one is TPR@1%FPR (true positive rate when the false positive rate is 1%) which was used by [4, 35] as a more practical measure since high FPR is not desirable for the attacker. Moreover, for WSA, we also consider the accuracy (ACC) of the attack model as a third metric, presented in Appendix A.

5.2. Evaluation of Proposed Attacks

Attack Performance Evaluation. We evaluate the attack performance of our proposed attacks, namely, CSA, AEA, and WSA, on target models. AEA utilizes widely-used transformations including resizing, crop, rotation, color jitter, translation, and horizontal flip.

From Table 1, we find that simple CSA already shows relatively high AUC scores (around 76% on LAION and 68% on CC12M). In addition, AEA consistently demonstrates slightly improved performance across all settings by 0.74% to 7%. While CSA and AEA demonstrate relatively high average performance, they fail in the low false-positive regime: the true positive rate is below 10% at the false positive rate of 1%. In contrast, WSA surpasses the performance of CSA and AEA in terms of all evaluation metrics by a significant margin. For instance, on the LAION dataset, WSA achieves an AUC of 93.2% and TPR@1%FPR of 74.01%, improving over CSA by 17% in terms of AUC and 64% in terms of TPR@1%FPR. Figure 2 illustrates examples that CSA fails to attack yet WSA succeeds. The effectiveness of WSA shows that there is a strong privacy implication of the one-sided non-member information available online. Overall, our results indicate a well-generalized multi-modal model trained with a large-scale dataset can still compromise privacy.

Interestingly, the attacks are less effective against models trained on CC12M than those trained on LAION. This is maybe because models trained on the CC12M dataset suffer more overfitting than models trained on the LAION



Figure 2. Examples of non-member samples that receive high cosine similarity scores (> 0.3). CSA cannot correctly mark these samples as non-members but WSA can.

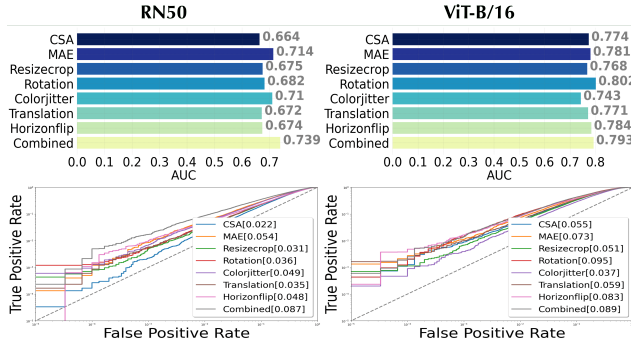


Figure 3. AEA performance with various augmentations on models pre-trained on two different datasets.

dataset. Moreover, given CC12M, the improvement induced by AEA is higher in RN50 and RN101 models compared with ViT-B/32. ViT-B/32 is overly large for the CC12M dataset, and the corresponding model suffers more overfitting than the other two ResNet-based architectures.

In-depth Analysis of AEA. We employ commonly utilized augmentations, such as resizing, crop, rotation, color jitter, translation, and horizontal flip, in conjunction with the masked autoencoder [11] to alter the feature space. In Figure 3, Combined denotes the AEA that utilizes all these augmentations. Combined is a reasonable strategy when the adversary does not have any knowledge to inform the selection of the augmentations.

Our experiments, shown in Figure 3, evaluate the effectiveness of AEA across different choices of augmentations. Firstly, Combined outperforms the CSA baseline in terms of the AUC and TPR@1%FPR metrics on both datasets, as demonstrated by the results provided in Table 1. Moreover, Combined exhibits the best or the second best performance compared to using individual transformations. This indicates that without further information guiding the selection of transformations, using the Combined approach is recommended.

In-depth Analysis of WSA. The cosine similarity score utilized by CSA can be regarded as a transformation of the image-text features but this transformation is too sim-

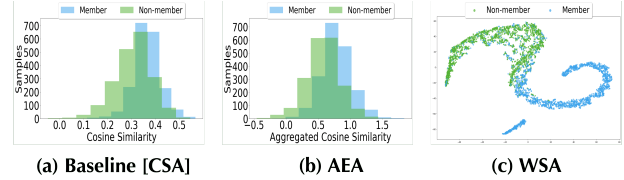


Figure 4. Comparison of the separability between members and non-members among CSA, AEA, and WSA.

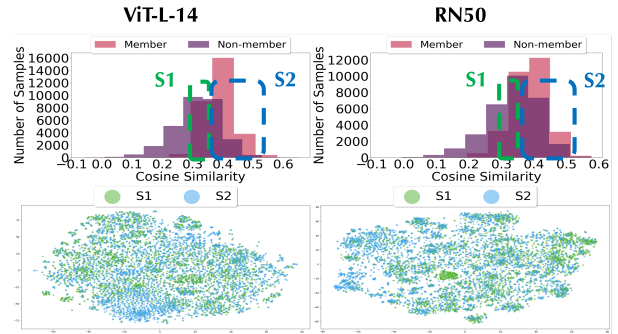


Figure 5. T-SNE plots for features from the pseudo-labeled (S2) and the overlapped area (S1), illustrating the discriminative feature alignment between the two areas.

ple to extract the membership-indicative information and map members and non-members to distinctive clusters. On the other hand, WSA trains an attack model, which can be regarded as a more complex transformation, thereby more successfully extracting the membership-indicative information from the image-text features of the target point. As shown in Figure 4, it is difficult to achieve a clear separation between members and non-members based on the CS score (the metric of CSA) or the aggregate CS score (the metric of AEA). Yet, they are separable in *some* latent space as shown in Figure 4 (c) which projects the member and non-member features onto a two-dimensional embedding space found by t-SEN [34]. The efficacy of WSA is primarily attributed to its active discovery of a latent space where member and non-member samples can be effectively separated.

Recall the pipeline of WSA. While the ground-truth non-member set has a diverse range of cosine similarity scores, the pseudo-members are labeled based on whether they

have higher cosine similarity than the majority of the non-members. In other words, the pseudo-members are mostly from the area $S2$ in Figure 5. On the other hand, the members used for evaluation have a broad cosine similarity range: some members have large enough cosine similarity, yet others are indistinguishable from non-members using cosine similarity, and WSA achieves a high performance overall (over 92% for all models trained with LAION and over 79% for all models trained with CC12M). An intriguing question thus arises: *Given that during training, the attack model has never seen members from the overlapped area (illustrated by $S1$ in Figure 5), how can the attack model learn to recognize these members?*

We hypothesize that the effectiveness of WSA in the overlapped area is due to **discriminative feature alignment** between non-overlapped and overlapped areas, i.e., the members from both areas contain similar features indicative of their membership. To verify the hypothesis, we randomly select data points whose CS scores lie in the overlapped area ($S1$) and conduct the same random sampling for the pseudo-labeled (or non-overlapped) area ($S2$). As shown in Figure 5, the features for members from the overlapped area have similar coverage to members from the non-overlapped area on both datasets. It is worth noting that features from the ViT-L/14 model, trained with the LAION dataset, exhibit better alignment compared to those from the RN50 model trained with CC12M. Specifically, the features extracted from the $S2$ area of the ViT-L/14 model span almost the entirety of the feature space; by contrast, the $S2$ features from RN50 are relatively sparse and concentrated and fail to cover the entire $S1$ area. The findings regarding feature alignment shed light on the superior attack performance, in terms of $\text{TPR@1\%FPR}$, achieved by the LAION-trained ViT-L/14 model compared to the CC12M-trained RN50 model, as shown in Table 1. The samples from the overlapped area ($S2$) can be considered as *hard-to-infer* samples, which contributes substantially to the $\text{TPR@1\%FPR}$, and discriminative feature alignment between $S1$ and $S2$ can increase the attack performance on *hard-to-infer* samples.

Runtime. Since our main focus is on attacking large-scale models, we provide the runtime analysis of each attack method. Table 1 shows the results. CSA and AEA take 670.7s and 830.1s respectively. WSA takes 1539.2s to perform the attack. This time requirement is almost negligible compared to existing shadow training-based methods, which involve retraining CLIP-scale models many times. The result shows our proposed attack techniques are efficient and scalable to large-scale models in practice.

5.3. Sensitivity Analysis

We will delve into two factors that could potentially affect attack performance: 1) the size of accessible non-members D_{no} and 2) the threshold for selecting pseudo-members.

Impact of $|D_{\text{no}}|$. We vary the size of D_{no} , ranging from 10K to 90K, and the results are summarized in Appendix C. The performance of WSA does vary with the size of accessible non-members yet only to a relatively small extent. In particular, it consistently outperforms the baseline for the range of $|D_{\text{no}}|$ considered.

Impact of Mislabeling Ratio of Pseudo-members. For WSA, we label data samples whose cosine similarities are higher than $\mu_{\text{no}} + \lambda\sigma_{\text{no}}$ as pseudo-members. Therefore, the selection of λ is crucial. We vary λ from -1.5 to 1.5 and evaluate our attack performance correspondingly in Table 2. We add another baseline that randomly selects samples from $\mathcal{D}_{\text{all}}$ and labels all the selected samples as members. This baseline does not rely on λ . Opting for a high threshold value enables us to collect cleaner training data samples; however, it may provide less information regarding samples that fall within the uncertain area (i.e., $S1$). Conversely, lowering the threshold may result in obtaining more information on the uncertain area, which could increase the $\text{TPR@1\%FPR}$, but would simultaneously reduce the attack model’s performance, as indicated in Table 2. Therefore, it is essential to balance both metrics to achieve reasonable performance. Our empirical findings suggest that $\lambda = 0.5$ yields reasonable performance. The performance drop at the very large threshold value of 1.5 is due to the insufficiency of pseudo-members; at the same time, there is little discriminative feature alignment between the non-overlapped area and the overlapped area. An in-depth analysis of Table 2 is provided in Appendix C.

6. Defenses

While defense against MIAs is not the primary focus of this work, we explore several well-established defense strategies for simple models, such as regularization [33], data augmentation, and differential privacy [28, 1, 16], and re-examine their effectiveness in the context of large-scale multi-modal learning.

L_2 Regularization. We choose the regularization hyperparameter $\alpha = 0.001$, as a higher value of α leads to non-convergence of loss. Our results are summarized in Table 3. We observe that WSA is resilient to the regularization-based defense in all metrics ($0.7754 \rightarrow 0.7463$ for AUC, $0.3016 \rightarrow 0.2748$ for $\text{TPR@1\%FPR}$, and $0.6779 \rightarrow 0.6567$ for ACC). AEA performance also drops by a little. However, the zero-shot performance on ImageNet deteriorates

Table 2. Attack performance vs. λ and corresponding mislabeled pseudo-member ratio. $\lambda = 0.5$ achieves the optimal performance.

Dataset [Model]	LAION [ViT-B-32]			LAION [ViT-L-14]			CC12M [ViT-B-32]			CC12M [RN50]		
Method	CSA	WSA	ACC	CSA	WSA	ACC	CSA	WSA	ACC	CSA	WSA	ACC
λ [Mislabel ratio]	TPR@1%FPR	TPR@1%FPR	ACC	TPR@1%FPR	TPR@1%FPR	ACC	TPR@1%FPR	TPR@1%FPR	ACC	TPR@1%FPR	TPR@1%FPR	ACC
Baseline [37.53%]	0.0671	0.6973	0.7390	0.0456	0.7072	0.7376	0.0400	0.2911	0.6475	0.0401	0.3132	0.6631
-1.5 [37.15%]	0.0694	0.6808	0.7419	0.0455	0.7178	0.7781	0.0440	0.2966	0.6559	0.0233	0.3640	0.6702
-0.5 [34.92%]	0.0688	0.7220	0.7338	0.0461	0.7050	0.7501	0.0412	0.3132	0.6698	0.0225	0.3703	0.6949
0 [32.87%]	0.0636	0.7106	0.8027	0.0475	0.7149	0.7675	0.0392	0.3000	0.6779	0.0264	0.3457	0.6856
0.5 [29.94%]	0.0653	0.6501	0.8081	0.0450	0.7199	0.8135	0.0419	0.2733	0.6966	0.0205	0.3036	0.7029
1.0 [26.43%]	0.0671	0.6304	0.7899	0.0448	0.6668	0.8199	0.0408	0.2604	0.7050	0.0269	0.2614	0.6625
1.5 [21.60%]	0.0623	0.5377	0.7810	0.0446	0.5628	0.8063	0.0378	0.2133	0.6757	0.0228	0.1947	0.6532

Table 3. Attack performance mitigation according to L2 regularization and data augmentation on ViT-B/32 model trained with CC12M.

ViT-B-32	Original				L2 [$\alpha = 0.001$]				DA			
Metric	AUC	TPR@1%FPR	ACC	Zeroshot	AUC	TPR@1%FPR	ACC	Zeroshot	AUC	TPR@1%FPR	ACC	Zeroshot
CSA	0.6952	0.0456	-		0.6874	0.0401	-		0.6589	0.0330	-	
AEA	0.7050	0.0775	-	21.67	0.6885	0.0689	-	16.87	0.6687	0.0515	-	22.87
WSA	0.7754	0.3016	0.6779		0.7463	0.2748	0.6567		0.7533	0.2583	0.6574	

Table 4. Attack performance mitigation according to the injection of various magnitudes of Gaussian noise.

	Model	ViT-B/32				ViT-B/16				RN50				RN101			
σ	Metric	AUC	TPR	ACC-L	VAL-L	AUC	TPR	ACC-L	VAL-L	AUC	TPR	ACC-L	VAL-L	AUC	TPR	ACC-L	VAL-L
$\sigma = 0.0$	CSA	0.7434	0.0632			0.7622	0.0506			0.6653	0.0226			0.6845	0.0300		
	AEA	0.7663	0.1056	0.6022	1.5306	0.7961	0.0925	0.6705	1.4910	0.7425	0.0893	0.3589	2.7325	0.7277	0.0880	0.2272	2.1100
	WSA	0.9252	0.6768			0.9352	0.6655			0.8058	0.3087			0.8325	0.3927		
$\sigma = 0.01$	CSA	0.7354	0.0593			0.7567	0.0518			0.6624	0.0171			0.6739	0.0329		
	AEA	0.7396	0.0901	0.6018	1.7223	0.7712	0.0853	0.6705	1.6410	0.7262	0.0780	0.3589	2.9098	0.215	0.0848	0.2274	2.1942
	WSA	0.9169	0.6510			0.9291	0.6677			0.8027	0.3138			0.8073	0.3421		
$\sigma = 0.5$	CSA	0.4979	0.0095			0.4991	0.0100			0.4962	0.0129			0.4976	0.0116		
	AEA	0.5038	0.0098	0.4055	1706.2	0.4997	0.0110	0.5006	1697.2	0.4999	0.0103	0.2867	2252.4	0.5035	0.0121	0.2218	865.43
	WSA	0.4795	0.0097			0.4893	0.0085			0.4938	0.0081			0.4958	0.0087		
$\sigma = 1.0$	CSA	0.4971	0.0093			0.4982	0.0104			0.5051	0.0101			0.5033	0.0100		
	AEA	0.5114	0.109	0.1204	6832.4	0.5054	0.0091	0.1711	6821.2	0.5050	0.0120	0.1534	9048.3	0.5134	0.0108	0.2094	3484.9
	WSA	0.4969	0.0087			0.4923	0.0098			0.4925	0.0109			0.4934	0.0094		

concurrently. Hence, the integration of L2 regularization mitigates the risk of privacy to a small extent but may potentially impede the utility.

Data Augmentation. In contrast, we find that data augmentation-based mitigation achieves a better tradeoff between privacy and utility. We leverage popular augmentations such as rotation, translation, and horizontal flip during training. The results in Table 3 show that the AUC score for WSA drops from 0.7754 to 0.7533 (TPR@1%FPR decreased by 0.0433), with an increase of 1.2% in zero-shot performance. We additionally provide the results on the RN50 model trained on CC12M with L2 and data augmentation in Appendix D.

Feature Perturbation. Because of the high computational costs incurred by modifying the CLIP training process, we consider a simple defense of adding noise into the output features of a pre-trained CLIP model, which is the in the same spirit as output perturbation approaches for classification models [17]. We inject zero-mean Gaussian noise

with different standard deviations into the features. The results are shown in Table 4. We include two evaluation metrics, namely, the ImageNet classification accuracy (i.e., ACC-I) and the validation loss (i.e., VAL-L). Increasing the Gaussian noise scale from $\sigma = 0$ to $\sigma = 1.0$ reduces attack performance to random guessing (i.e., 0.5). In particular, until $\sigma = 0.01$, WSA, AEA, and CSA show relatively high attack performance. Hence, if we want to defend against WSA, noise with at least $\sigma = 0.5$ is needed; however, at the same time, it causes a significant increase in validation loss and degradation in zero-shot performance (e.g., from 0.60 to 0.40 for ViT-B/32).

Differential Privacy (DP). DP offers a formal guarantee for protecting individual records, which is suitable for defending against MIAs. We implement the DP with the Opacus library [38]. To reduce the dependency among gradients within a batch, we replace the batch normalization layer with an alternative normalization as suggested in Opacus. However, this substitution has already led to convergence issues of CLIP (see Appendix D). Furthermore, training a

large-scale model incorporating differential privacy leads to resource limitation issues. Therefore, incorporating DP into CLIP necessitates significant additional efforts to re-design the contrastive training process to be DP-friendly (i.e., reducing the dependency between individual gradients) and computationally efficient. We will leave it as an open problem.

7. Conclusion and Outlook

This paper presents the first focused study on application-agnostic MIAs against large-scale multi-modal models. To that end, we develop three strategies that make MIAs practical for large-scale models. Our evaluation shows the efficacy of the proposed attack strategies. Furthermore, our experimental findings emphasize the importance of safeguarding privacy for multi-modal foundational models, which were previously believed to be less susceptible to overfitting.

Our work opens several exciting directions for future work. First, it will be interesting to adapt our attack method to other foundation models, like BERT and GPT-3. Second, while the injection of Gaussian noise provides a simple defense, it does not come with provable privacy guarantees. How to build high-utility foundation models with rigorous privacy guarantees is an interesting open question.

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